

Indo-American DEFENCE INNOVATION INDEX

IADI | Index Methodology & Rulebook

1. THEME & RATIONALE

The IADI tracks publicly listed companies across the United States and India that derive a meaningful portion of their revenue from defence manufacturing, AI-enabled autonomous systems, and cybersecurity / electronic warfare. The index captures the intersection of three structural trends:

- Geopolitical realignment:** Record defence budgets globally post-2022, with India targeting self-reliance under Aatmanirbhar Bharat and the US sustaining multi-theatre deterrence spending.
- AI in warfare:** Autonomous systems, drone swarms, and AI-enabled command-and-control are reshaping procurement priorities across both nations.
- Gap vs. existing indices:** Existing ETFs (ITA, XAR, DFEN) are US-only. No bilateral US-India defence index exists. IADI fills this white space with a risk-optimised, rules-based approach denominated in INR for Indian investors.

2. ELIGIBILITY CRITERIA

Criterion	Rule
Exchange Listing	Listed on NYSE, NASDAQ (US) or NSE, BSE (India)
Sub-theme	Must operate in ≥1 of: (i) AI & autonomous systems / drones, (ii) defence manufacturing & platforms, (iii) cybersecurity & electronic warfare
Revenue Exposure	≥33% of trailing 12-month revenue attributable to defence / aerospace contracts or programmes
Market Cap	US: Free-float market cap ≥ USD 500M India: Free-float market cap ≥ ₹1,000 Cr at reconstitution
Share Class	Primary listings only - ADRs and GDRs excluded to avoid double-counting
Currency	Index denominated in INR. US constituents converted at daily USD/INR close rate (USD/INR=X). Indian investors bear FX risk on the US allocation.

3. LIQUIDITY FILTERS

Minimum 3-month ADTV computed over 63 trading days prior to each reconstitution:

US: ADTV ≥ USD 10 million

India: ADTV ≥ ₹15 Crore (INR 150M)

ADTV computed as $\text{mean}(\text{close} \times \text{volume})$ in native currency over the 63-day window.

Buffer rule: Existing constituents apply 20% buffer on ADTV threshold to reduce unnecessary churn.

4. NUMBER OF CONSTITUENTS

Constituent count is not fixed. After applying eligibility and liquidity filters, the portfolio size is determined by a Mean-Variance Minimum Volatility optimisation (Stage A) that minimises ex-ante portfolio variance subject to:

- Individual weight ≥ 2% and ≤ 10%
- Weights sum to 100%
- No leverage or short positions
- Covariance computed on trading-day intersection with ridge regularisation ($\lambda = 1 \times 10^{-6}$)

Expected range: 20–40 stocks

5. WEIGHTING METHODOLOGY

v1.1 UPDATE — CHANGED FROM v1.0

v1.0 method: Inverse Volatility $w_i = (1/\sigma_i) / \sum (1/\sigma_j)$

v1.1 method: Market Cap Weighting $w_i = \text{mcap}_i / \sum \text{mcap}_j$, capped at 5%

Rationale: Systematic comparison of 6 weighting methodologies (Inverse Volatility, Equal Weight, Market Cap, Minimum Variance, Maximum Diversification, Risk Parity) on 3+ years of backtest data showed Market Cap produced the highest Sharpe ratio and the best return. In the Indian defence re-rating of 2022–2024, size was the dominant signal - larger names (HAL, BEL, MAZDOCK) drove bulk sector returns. Inverse-vol overweighted smaller, lower-vol names that underperformed.

Stage A — Constituent Selection

Mean-Variance Minimum Volatility QP selects the constituent set. Stocks awarded non-trivial weight in the optimiser output form the index. This stage uses the full covariance structure to filter names that would add portfolio variance.

Stage B — Weight Assignment (v1.1)

Free-float market cap weighted allocation within the selected constituent set. Each stock's weight is proportional to its market capitalisation, capped at 5% per constituent, floored at 2%, normalised to sum to 100%.

Formula: $w_i = \text{mcap}_i / \sum \text{mcap}_j$, 5% cap, 2% floor

6. REBALANCING & RECONSTITUTION

Event	v1.0 Schedule	v1.1 Schedule
Weight Rebalance	Quarterly: last trading day of Mar, Jun, Sep, Dec	Quarterly: last trading day of Mar, Jun, Sep, Dec (unchanged)
Reconstitution	Annual: first trading day of January	Semi-annual: first trading day of January AND July
Buffer Rule	20% buffer on ADTV threshold for existing constituents	Unchanged

Rationale for semi-annual reconstitution: The Indian defence universe evolved rapidly in 2022–2024. New listings (DCX Systems, Data Patterns, Cochin Shipyard) and fast-rising mcap names entered eligibility within months. Annual reconstitution was too slow to capture these names; semi-annual review improved all key metrics (total return, Sharpe, max drawdown). Between rebalance dates, weights drift naturally with price movements.

7. BENCHMARK & DATA

Benchmark	Composite: 50% ITA (iShares US Aerospace & Defence ETF, INR-converted) + 50% Nifty India Defence Index (reconstructed from official NSE methodology). Reflects bilateral investment universe.
Nifty Defence	Reconstructed from 7 semi-annual constituent snapshots (Jan 2022 – Jan 2025) using official NSE methodology: free-float mcap weighted, 20% stock cap, semi-annual reconstitution. Data source unavailable via free providers; reconstruction documented in decision log.
Base Value	1,000 Base Date: 3 January 2022
Price Data	Adjusted close prices (Yahoo Finance via yfinance). Corporate actions (splits, dividends) reflected automatically.
Currency	INR throughout. US constituents converted at daily USDINR=X close rate. Benchmark ITA leg likewise converted to INR for apples-to-apples comparison.
Risk-Free Rate	6.5% p.a. (Indian 10-yr Government Securities yield, INR-denominated index). Used for Sharpe ratio calculation.

8. METHODOLOGY EVOLUTION (V1.0 → V1.1)

The following table summarises all changes between Rulebook v1.0 (January 2022) and v1.1. Changes were determined by systematic empirical analysis including sensitivity analysis and a six-method weighting comparison over the full 3-year backtest period.

Parameter	v1.0	v1.1 (Updated)
Weighting Method	Inverse Volatility (126-day)	Market Cap (free-float)
Single Stock Cap	5% maximum	10% maximum
Single Stock Floor	2% minimum	2% minimum (unchanged)
Reconstitution	Annual (January)	Semi-annual (Jan + Jul)
Weight Rebalance	Quarterly	Quarterly (unchanged)
Theme / Eligibility / Liquidity / Benchmark	As per v1.0	Unchanged